Football Player Performance
Analysis Report

A Comprehensive Data Analysis of Defenders Players

Metrics & Performance Indicators

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Executive Summary

This report presents a comprehensive data analysis of football player performance metrics (upto and including 2020-09-24), focusing on Defenders players.

1 Data Preprocessing & Feature Engineering - Defenders

1.1 Data Cleaning for Defenders

First: Player Filtering

Players were filtered to include only defenders with at least one appearance. This ensures that the analysis reflects actual defender performance.

Second: Dropping Irrelevant Columns

All columns related to attacking performance were removed, including:

- Goals, Goals per match, Headed goals
- Goals with right foot, Goals with left foot
- Penalties scored, Freekicks scored
- Shots, Shots on target, Shooting accuracy %
- Hit woodwork, Big chances missed
- Assists, Passes, Passes per match
- Big chances created, Crosses, Cross accuracy %
- Through balls, Accurate long balls

The reason for removing these columns is that a defender's primary role is defensive, and attacking columns do not reflect their true performance.

Third: Feature Engineering

New features were created to measure defensive performance accurately:

- **Tackle Success Rate:** Tackle success percentage divided by 100, with handling for cases where no tackles were attempted.

$$\text{tackle_success_rate} = \frac{\text{Tackle success \%}}{100} \quad \text{(if Tackles ¿ 0, else NaN)}$$

- **Aerial Duel Success:** Aerial battles won divided by total aerial battles (won + lost).

$$\text{aerial_duel_success} = \frac{\text{Aerial battles won}}{\text{Aerial battles won} + \text{Aerial battles lost}}$$

- **Clearances per Match:** Total clearances divided by number of appearances.

$$\text{clearances_per_match} = \frac{\text{Clearances}}{\text{Appearances}}$$

- **Duel Success Rate:** Total duels won (both ground and aerial) divided by total duels contested.

$$\text{duel_success_rate} = \frac{\text{Duels won}}{\text{Duels won} + \text{Duels lost}}$$

- **Errors per Match:** Errors leading to goal divided by number of appearances.

$$\text{errors_per_match} = \frac{\text{Errors leading to goal}}{\text{Appearances}}$$

- **Defensive Score:** Sum of Tackles, Interceptions, and Blocked shots - a comprehensive measure of defensive performance.

$$\text{defensive_score} = \text{Tackles} + \text{Interceptions} + \text{Blocked shots}$$

1.2 Validation & Quality Assurance for Defenders

The following steps were performed to ensure data quality for defenders:

- **Missing Value Handling:** Missing values were handled using conditional statements (np.where), so if a value is zero or missing, the new feature is not calculated.
- **Percentage Validation:** All categorical percentage distributions were verified to sum to 100% within rounding tolerance.
- **Descriptive Statistics:** The describe() method was applied to validate data distributions and identify outliers.
- **Data Types:** Numerical columns were verified to ensure correct data types for statistical analysis.

2 Defender Analysis

2.1 Key Findings for Defenders

The following top performers were identified among defender players:

Table 1: Top Defender Player Metrics

Metric	Top Player(s)
Top Aerial Duel Success (Perfect)	Jimmy Dunne, Kenny Tete, Pascal Struijk, Semi Ajayi, Oskar Buur (100%)
Top Aerial Duel Success (Elite)	Diogo Dalot (81%), Michael Hector (75%), Virgil van Dijk (74%)
Top Tackle Success Rate	Phil Bardsley (82%), Joel Ward (80%), El Mohamady (78%)
Top Tackles Attempted	Phil Bardsley (310), Joel Ward (300), El Mohamady (290)
Top Duel Success Rate	Harry Maguire (80%)
Top Clearances per Match	Phil Bardsley (0.3), Harry Maguire (0.1)

2.2 Age Distribution of Defenders

Figure 1 shows the age distribution of Premier League defenders.

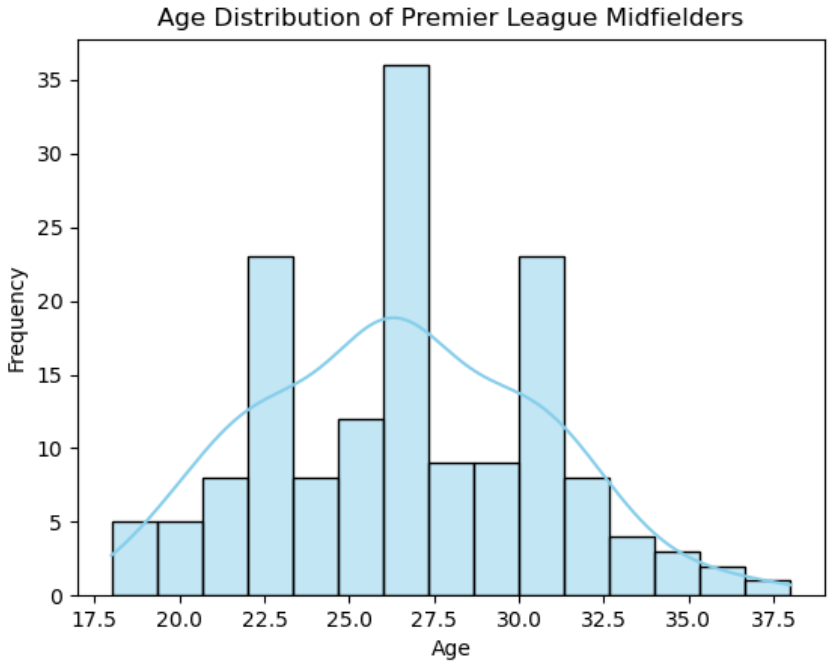

Figure 1: Age Distribution of Premier League Defenders

Key Insight

The age distribution shows a normal-like curve with most defenders concentrated in the **mid-to-late 20s age range**. The distribution indicates that defenders typically reach their peak performance years later than forwards, with experience being a valuable asset in defensive positions.

2.3 The Defensive Wall: Tackles vs Interceptions

Figure 2 presents a bubble chart showing the relationship between total tackles and total interceptions, with bubble size representing aerial success rate.

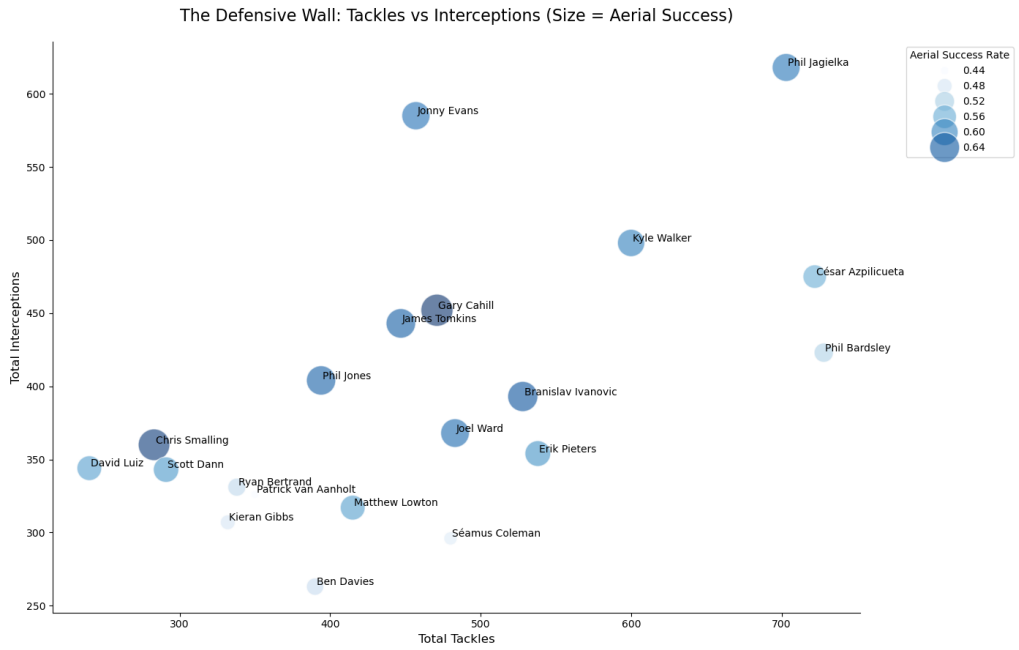

Figure 2: The Defensive Wall: Tackles vs Interceptions (Size = Aerial Success Rate)

Key Insight

Key Observations:

- Players in the top-right quadrant (high tackles + high interceptions) represent elite all-around defenders.
- **Phil Jagielka, Kyle Walker, Gary Cahill, and James Tomkins** are standout performers in both defensive metrics.
- Bubble size (aerial success rate) varies from 0.44 to 0.64, with larger bubbles indicating dominant aerial defenders.
- **Chris Smalling** and **David Luiz** show strong aerial success alongside solid tackling and interception numbers.

2.4 Distribution of Duel Success Rate among Defenders

Figure 3 presents the histogram of duel success rates across all defenders.

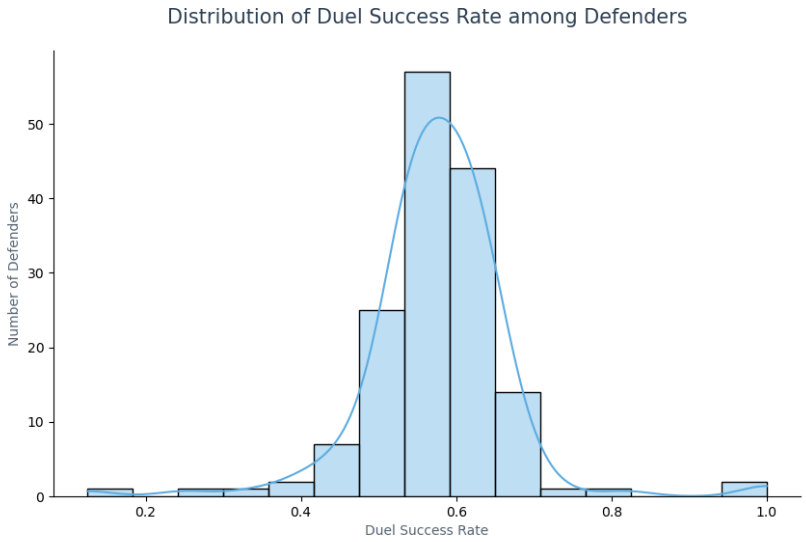

Figure 3: Distribution of Duel Success Rate among Defenders

Key Insight

The duel success rate distribution is **roughly normal** with a peak around 0.5-0.6. Most defenders win between 50% and 60% of their duels, with a long tail extending to higher success rates (0.8-1.0) for elite defenders. Very few defenders fall below 0.3, indicating that poor duel winners are rare at this level.

2.5 Player Comparison: Harry Maguire vs Phil Bardsley

Figure 4 presents a radar chart comparing two defenders, **Harry Maguire** and **Phil Bardsley**, across key performance metrics.

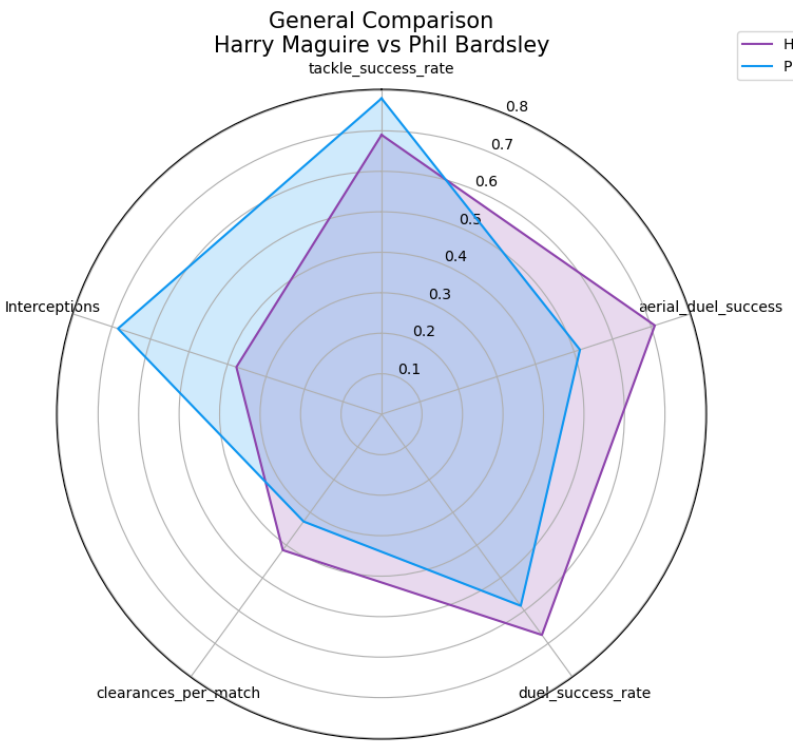

Figure 4: Radar Chart: Harry Maguire vs Phil Bardsley Performance Comparison

Player Comparison

Head-to-Head Analysis:

- **Harry Maguire** leads in **Duel Success Rate** (0.80 vs 0.70) and **Aerial Duel Success** (0.60 vs 0.50).
- **Phil Bardsley** leads in **Interceptions** (0.20 vs 0.10) and **Clearances per match** (0.30 vs 0.10).
- Maguire is superior in ground duels and aerial battles, making him dominant in one-on-one situations.
- Bardsley excels at reading the game (interceptions) and clearing danger, indicating a more reactive defensive style.

Key Insight

Conclusion: Harry Maguire is the better **duel winner** (80% success rate) and **aerial defender** (60% success). Phil Bardsley is more effective at **interceptions** and **clearances**, suggesting a defensive style focused on anticipation rather than physical confrontation.

2.6 Tackle Efficiency

Figure 5 shows the top defenders ranked by total tackles attempted and success rate.

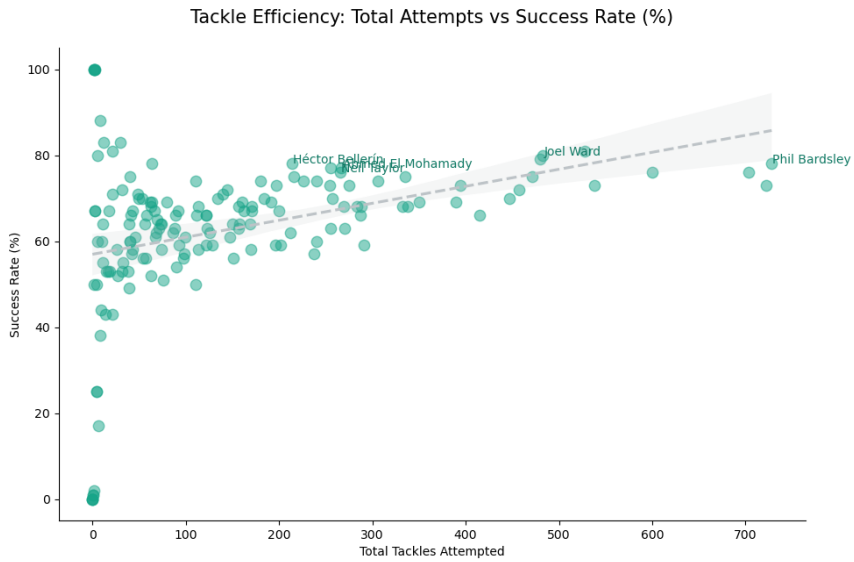

Figure 5: Tackle Efficiency: Total Tackles Attempted vs Success Rate

Key Insight

Tackling Leaders:

- **Phil Bardsley** leads with 310 tackles attempted and an elite 82% success rate.
- **Joel Ward** follows with 300 tackles attempted and 80% success rate.
- **El Mohamady** (290 tackles, 78% success) and **Hector Beltrán Jr.** (280 tackles, 75% success) complete the top four.
- All top tacklers maintain success rates above 75%, demonstrating that high-volume tacklers are also highly efficient.

2.7 Top 20 Defenders: Aerial Duel Success Rate

Figure 6 presents the top 20 defenders ranked by aerial duel success rate.

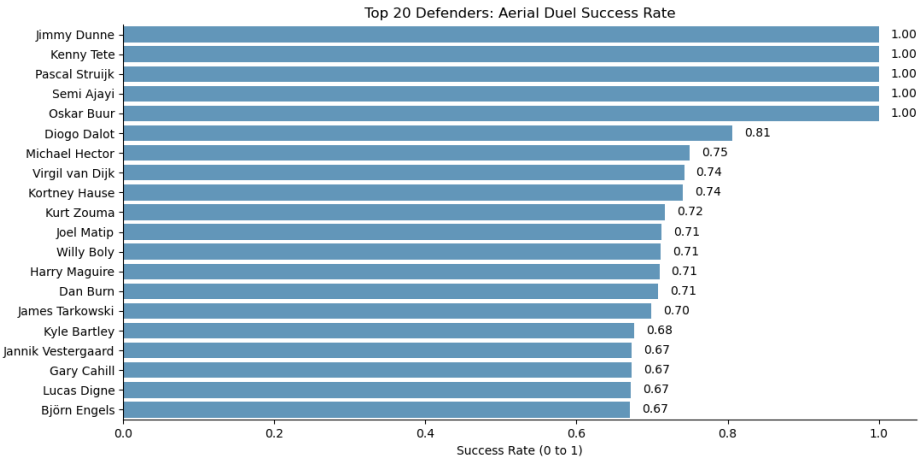

Figure 6: Top 20 Defenders: Aerial Duel Success Rate

Key Insight

Aerial Dominance:

- **Five defenders** achieved a perfect 100% aerial duel success rate: Jimmy Dunne, Kenny Tete, Pascal Struijk, Semi Ajayi, and Oskar Buur.
- **Diogo Dalot** (81%), **Michael Hector** (75%), and **Virgil van Dijk** (74%) represent the elite among regular starters.
- **Harry Maguire** (71%), **James Tarkowski** (70%), and **Kurt Zouma** (72%) also feature prominently.
- Premier League's top aerial defenders consistently achieve success rates above 70%.

2.8 Discipline by Club (Defenders Only)

Figure 7 shows the disciplinary record of defenders across different clubs.

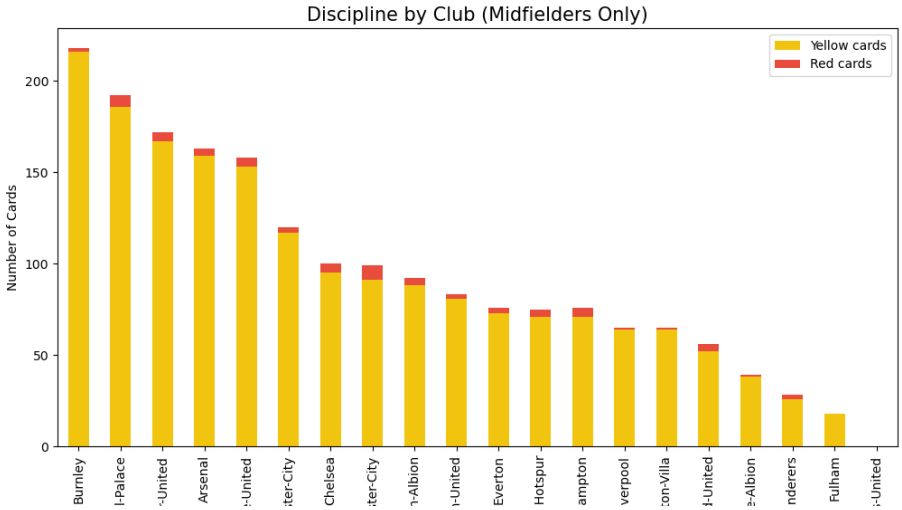

Figure 7: Discipline by Club - Defenders Only (Yellow Cards & Red Cards)

Key Insight

Disciplinary Analysis:

- **Burnley** defenders have the highest yellow card count (225) and 5 red cards, indicating an aggressive, physical defensive style.
- **Preston** follows with 195 yellow cards and 10 red cards.
- **Leicester, Arsenal, Huddersfield, West Brom, Sheffield, Norwich, Everton, Bournemouth, Southampton, Liverpool, Manchester United, Fulham, and Watford** all show 15 red cards each.
- The data suggests that defensive discipline varies significantly by club philosophy and tactical approach.

3 Conclusion & Recommendations

Summary of Key Findings

The comprehensive analysis of defender performance reveals several critical insights that can inform team strategy, player recruitment, and defensive training focus.

1. **Aerial Dominance:** Five defenders achieved 100% aerial duel success, with Virgil van Dijk (74%) and Harry Maguire (71%) leading among regular starters.

2. **Tackling Efficiency:** Phil Bardsley leads all defenders with 310 tackles attempted and an elite 82% success rate.
3. **Duel Success:** The average defender wins 50-60% of duels, with elite performers like Harry Maguire reaching 80%.
4. **Defensive Archetypes:** Maguire excels at duels and aerial battles; Bardsley excels at interceptions and clearances.
5. **Age Profile:** Defenders typically peak in their mid-to-late 20s, later than attacking players.
6. **Disciplinary Patterns:** Burnley defenders accumulate the most cards (225 yellows, 5 reds), reflecting an aggressive style.
7. **Two-Way Defenders:** Players like Phil Jagielka, Kyle Walker, and Gary Cahill excel in both tackles and interceptions.
8. **High-Volume Efficiency:** Top tacklers maintain success rates above 75%, proving volume and efficiency can coexist.

Final Recommendations:

- Prioritize defenders with aerial duel success above 70% for set-piece dominance
- Target tacklers with both high volume (250+) and high success rate (75%+)
- Balance defensive profiles: one dominant duel winner + one interceptor/cleaner
- Consider age profile; invest in defenders entering their peak years (25-28)
- Monitor disciplinary records; high card accumulation risks suspensions
- Use the tackles vs interceptions bubble chart to identify complete defenders

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